

SMAART Minds Assessment System & Technical Report

Version: 1.0

Date: February 3, 2026

Reference Document: SMAART Minds Complete User Journey (v1.0)

1. Executive Summary

This document provides a comprehensive technical overview of the SMAART Minds assessment ecosystem, specifically focusing on the **T1 Baseline Assessment** logic, scoring algorithms, system architecture ("the build"), and reporting mechanisms. It serves as the definitive guide to how the system measures and classifies student capabilities.

2. Technical Building Blocks ("The Build")

The system is built on a modern, scalable MERN stack architecture designed for performance and security.

2.1 Backend Architecture

- **Runtime Environment:** Node.js with Express.js framework.
- **Database:** MongoDB (using Mongoose ODM) for flexible schema modeling of complex assessment results.
- **Authentication:**
 - JWT (JSON Web Tokens) for stateless session management.
 - OTP-based 2FA for secure login.
 - Bcrypt encryption for password hashing.
- **API Structure:** RESTful API endpoints handling:
 - User Profiles (/api/users)
 - Assessment Delivery (/api/developments , /api/results)
 - Course Management (/api/courses)

2.2 Frontend Architecture

- **Framework:** React 18.3 with Vite 5.4 for high-performance building.
- **UI System:** Radix UI primitives with Tailwind CSS for styling.
- **State Management:** TanStack Query (React Query) for efficient server state

synchronization.

- **Routing:** React Router DOM (v6) for client-side navigation.

3. T1 Baseline Assessment Logic

The T1 Baseline Assessment is the core diagnostic tool used to establish a student's starting profile.

3.1 Assessment Structure

- **Assessment Code:** ASM00001
- **Total Questions:** 36 Questions
- **Format:** Multiple Choice (4 options, 1 correct answer)
- **Time Limit:** Untimed (but enforces minimum 5-second reflection time per question)

3.2 The 6 Quotients

The 36 questions are stratified across 6 key quotients (dimensions):

Quotient	Full Name	Questions	Focus Area
CRQ	Cognitive Readiness Quotient	7	Critical thinking, logic, problem-solving
SRQ	Self-Regulation & Drive Quotient	6	Motivation, emotional control, discipline
LQ	Learning Agility Quotient	6	Adaptability, continuous learning mindset
SIQ	Social Interaction Quotient	6	Collaboration, empathy, teamwork
PEQ	Professional Execution Quotient	7	Work ethic, reliability, delivery
DAQ	Digital & AI Literacy Quotient	4	Tech proficiency, AI readiness

3.3 Question Assignment

- Questions are fetched from the database using a **Stratified Sampling** algorithm.
- The system ensures the exact distribution of questions per quotient (e.g., exactly 7 CRQ questions, 6 SRQ questions) is presented to every user to ensure standardization.

4. Scoring Logic & Algorithms

The scoring engine processes results in real-time upon submission of each answer.

4.1 Question Scoring

Each question uses binary scoring:

```
// Logic
const score = (selectedAnswer === correctAnswer) ? 1 : 0;
```

- **Correct Answer:** +1 point
- **Incorrect Answer:** 0 points

4.2 Quotient Scoring

For each of the 6 quotients, a percentage score is calculated:

1. **Count Correct:** Sum of correct answers for that quotient.
2. **Calculate Percentage:** $(\text{Correct} / \text{Total Questions in Quotient}) * 100$

Example (CRQ):

- Student answers 5 out of 7 questions correctly.
- Score: $(5 / 7) * 100 = 71.4\%$ (Rounded to 71%)

4.3 Band Classification

Every percentage score is mapped to a specific proficiency band:

Score Range	Band	Classification
81% - 100%	 Advanced	Exceptional mastery
61% - 80%	 Strong	Solid competence
41% - 60%	 Progressing	Developing skills
21% - 40%	 Developing	Early stage
0% - 20%	 Emerging	Beginning journey

4.4 Baseline Score (S_baseline)

The overall "T1 Score" is not just a sum of correct answers, but a weighted average of the quotient performances to ensure balanced representation.

```
BaselineScore = Average(CRQ%, SRQ%, LQ%, SIQ%, PEQ%, DAQ%)
```

5. Reporting & Outputs

5.1 Database Storage

Results are stored in the `BaseLineResult` collection with the following schema structure:

```
{
  "userId": "ObjectId(...)",
  "baselineScore": 54,           // The overall average score
  "stageBand": "Progressing",  // The overall band
  "t1Profile": {
    "CRQ": { "rawScore": 71, "level": "Strong", "earned": 5, "possible": 7 },
    "SRQ": { ... },
    // ... other quotients
  },
  "totalScore": 24,             // Total correct answers (out of 36)
  "createdAt": "Timestamp"
}
```

5.2 User Report (Downloadable)

The system generates a detailed text-based report (`FinalMinds_T1_Baseline_[ID].txt`) containing:

1. **Student Metadata:** Name, ID, Date.
2. **Readiness Index:** The calculated Baseline Score and Band.
3. **Quotient Breakdown:** Detailed performance in each of the 6 areas.

4. **Strategic Recommendations:** Automated advice based on the identified "Emerging" or "Developing" areas.
-

6. Access Control & Proctoring

To ensure data integrity, the assessment engine includes:

- **Right-Click Disable:** Prevents copying questions.
- **Tab Switch Detection:** Logs warnings if the user leaves the assessment window.
- **Auto-Submit:** Forces submission after 3 suspicious activities (warnings).